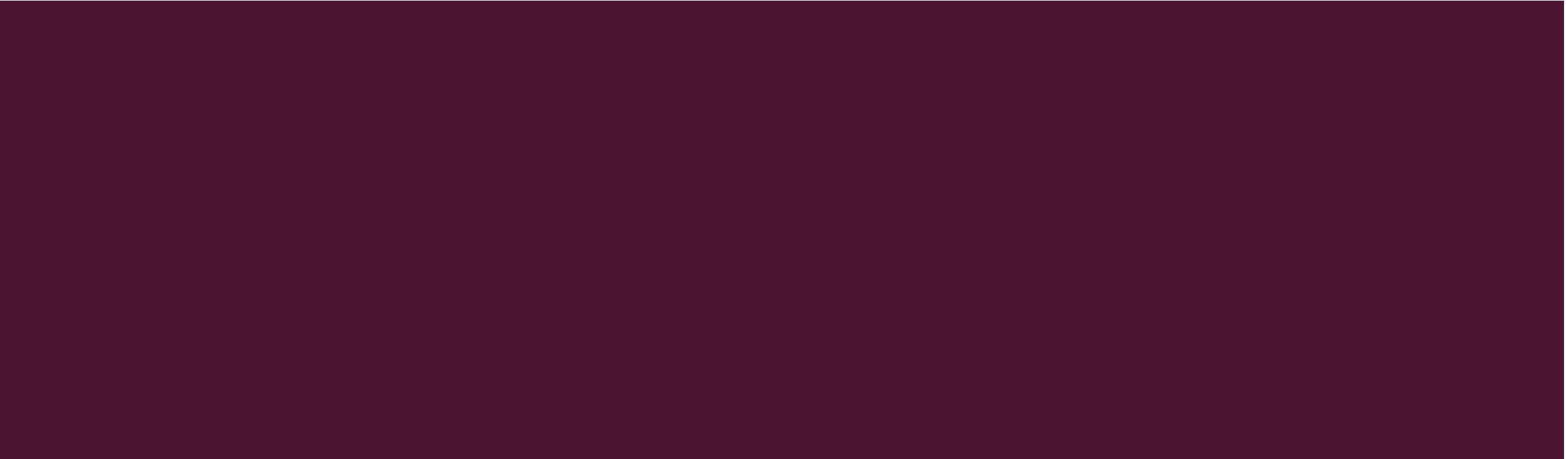

CHAPTER 3

CRYPTOGRAPHIC CONCEPTS & CLASSICAL CRYPTOGRAPHY





Cryptography

"The Art of Secret Writing"

Mathematics
Protects Secrets,
Not Hiding.

Openly Known.
Still Secure.

The Meaning of Cryptography

Cryptography comes from the Greek words **kryptos**, meaning "concealed, hidden, veiled, secret, or mysterious" and **graphia**, meaning "writing". Thus, cryptography is **"the art of secret writing"**.

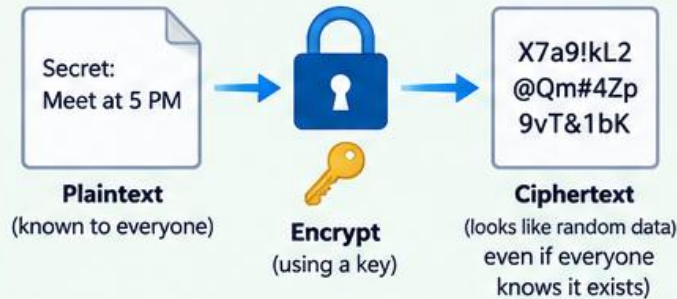


Cryptography vs. Security Through Obscurity



Cryptography

Security through mathematics



With cryptography, it does not matter if third-parties know of the existence of the secret, because they can never know what it is without obtaining an **appropriate credential** (the key).



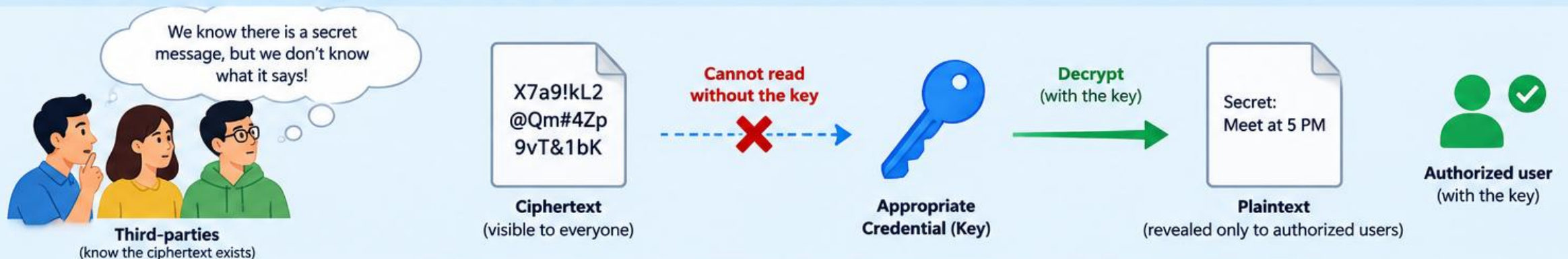
Security Through Obscurity

Keeping something a secret by hiding it



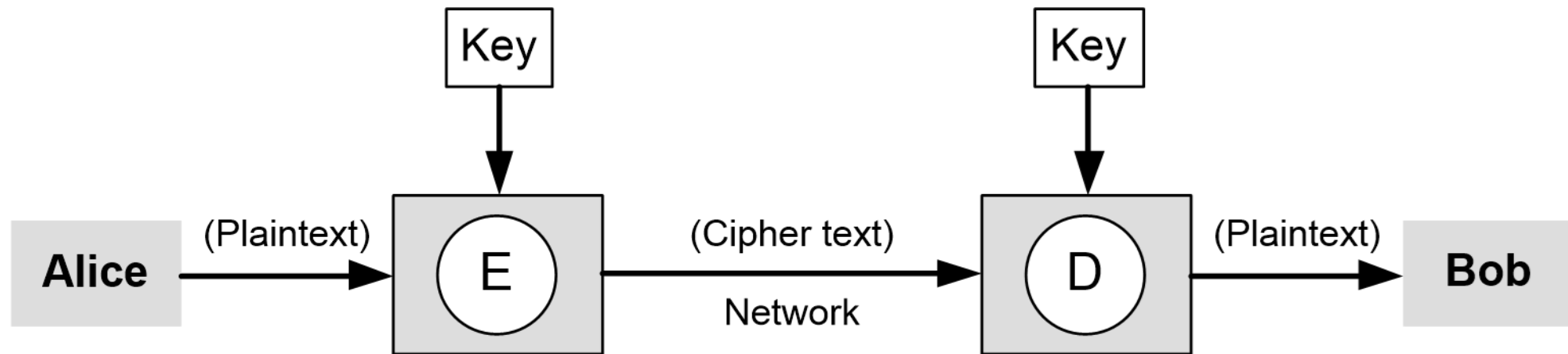
Relies on **hiding the existence** of the information. If someone discovers it, the secret is lost.

Even If Others Know, It's Still Secure



CRYPTOGRAPHY (CONT.)

- **Encryption** uses a *reversible process* based on a secret
 - The use of a **key** with the **encryption cipher** (aka **encryption algorithm**) ensures that decryption can only be performed by authorized persons.



Encryption Algorithms (Ciphers)

Ciphers are **algorithms** used to encrypt or decrypt the data.

Same Algorithm
Encrypts and Decrypts
a Message (with key)



Data
Protected
Information
Safer World



Types of Ciphers



Classical Ciphers

Early encryption techniques based on manipulating the letters in plaintext.

1 Substitution Cipher

A block of plaintext is replaced with ciphertext.

Plaintext

H E L L O

Substitution
Algorithm

Ciphertext

X E K K C

Each letter (or block) is replaced with another letter (or block).

2 Transposition Cipher

The letters of the plaintext are shifted about to form the cryptogram.

Plaintext

H E L L O

Transposition
Algorithm

Ciphertext

L O H E L

The letters are rearranged, but not replaced.



Modern Ciphers

Modern encryption algorithms are classified based on the key used and the type of input data.



Based on the type of key used



3 Private Key (Symmetric Cipher)

Same key is used for encryption and decryption.



Plaintext



Same key
(encrypt & decrypt)



Ciphertext



4 Public Key (Asymmetric Cipher)

Two different keys are used for encryption and decryption.



Plaintext



Public key
(encrypt)



Private key
(decrypt)



Ciphertext



Based on the type of input data



5 Block Cipher

Encrypts blocks of data of fixed size.

1010 1100 0110 1001

Fixed-size blocks



6 Stream Cipher

Encrypts continuous streams of data.

0101100101010100011...

Continuous stream of data

CLASSICAL CIPHERS

Classical Encryption Techniques

- Classical encryption can be categorized into several types, each employing different techniques for generating secret codes, as illustrated by the following examples.

Substitution Technique	Transposition Technique
- Monoalphabetic Substitution → Atbash → Caesar Cipher → Pigpen Cipher	- Simple Transposition Cipher - Rail Fence - Simple Columnar Transposition - Double Transposition Cipher
- Polygraphic Substitution → Playfair Cipher	
- Polyalphabetic Substitution → Vigenere Cipher → Vernam Cipher	

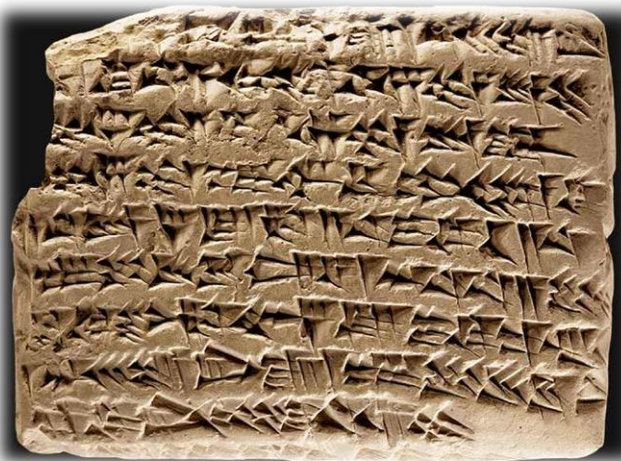
ANCIENT/HISTORICAL ENCRYPTION (CONT.)



3200-3500 BC: Sumerians

■ Cuneiform writings

- the oldest known writing system, created in ancient Mesopotamia
- The word comes from Latin *cuneus*, meaning "**wedge-shaped**," because of the wedge marks made in clay.
- **Not a language:** It is a script or writing system used to write many languages, such as Sumerian, Akkadian, and Hittite.

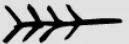


ANCIENT/HISTORICAL ENCRYPTION (CONT.)

Cuneiform: From pictograph to Assyrian

original pictograph	pictograph in position of later cuneiform	early Babylonian	Assyrian	original or derived meaning
				bird
				fish
				donkey
				ox
				sun, day

<https://www.britannica.com/topic/writing/Sumerian-writing>
<https://www.omniglot.com/writing/sumerian.htm>

				grain
				orchard
				to plow, to till
				boomerang, to throw, to throw down
				to stand, to go

Courtesy of the Oriental Institute, the University of Chicago.

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

1900 BC: Old Kingdom of Egypt

- **Hieroglyphics symbols**

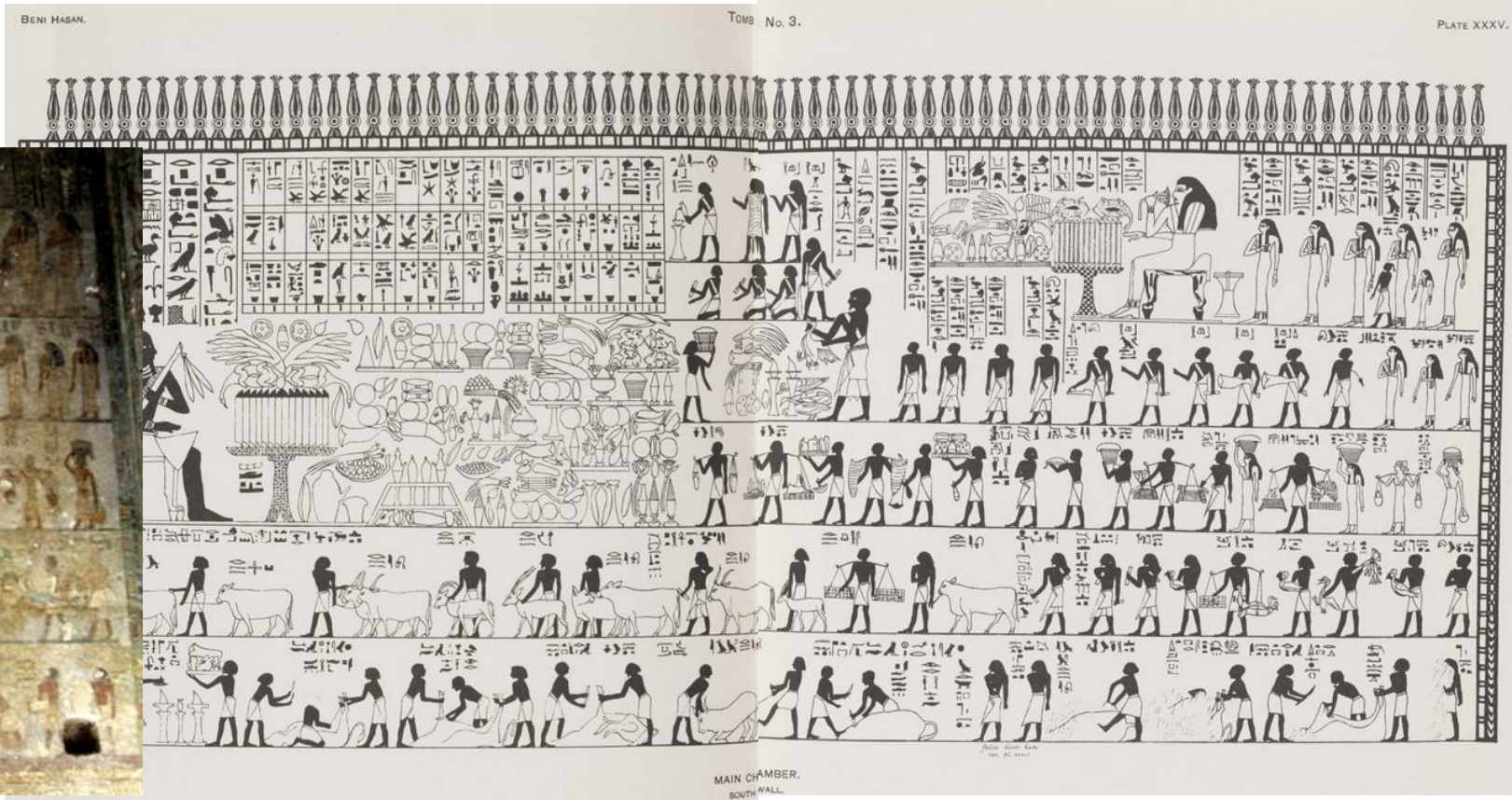
- First known use of cryptography
- An inscription carved into the main chamber of the tomb of nobleman Khnumhotep II



ANCIENT/HISTORICAL ENCRYPTION (CONT.)

1900 BC: Old Kingdom of Egypt

- Hieroglyphics symbols



https://en.wikipedia.org/wiki/Khnumhotep_II

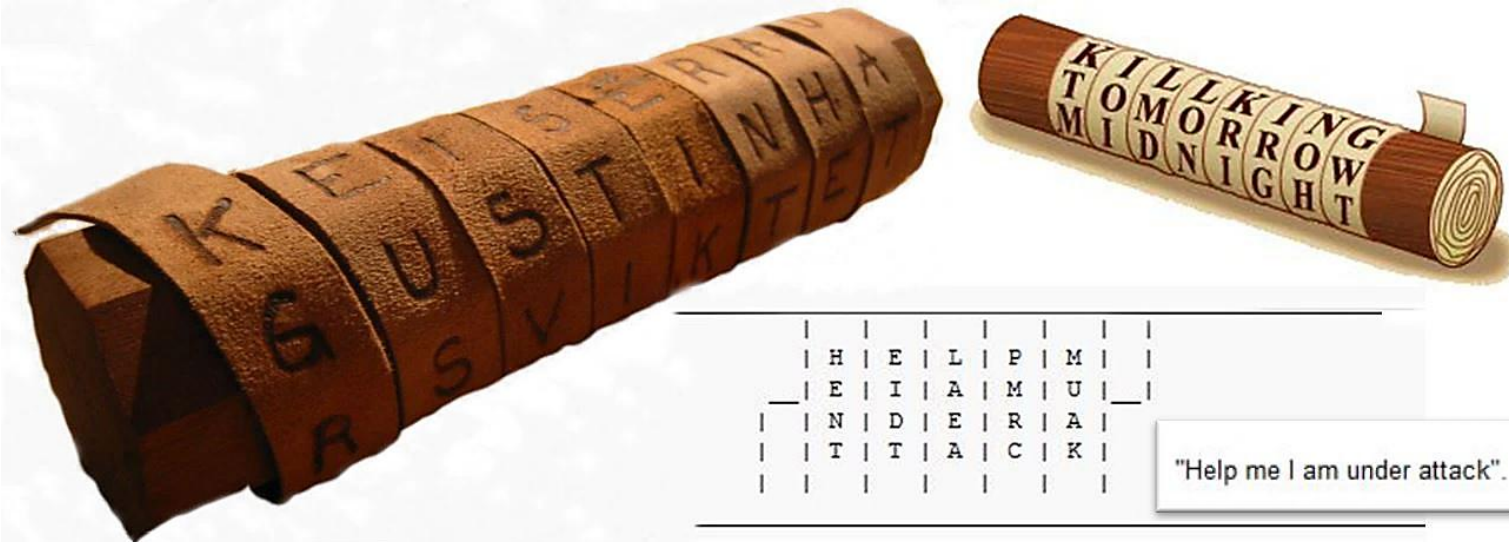
ANCIENT/HISTORICAL ENCRYPTION (CONT.)



650 BC: Ancient Greeks

■ Scytale Cipher

- Used by the Spartan military
- Wrapping a leather strip around a wooden rod (scytale) of a specific diameter to read the message



AAAATCTWTKDN

Can you decrypt it?

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

500-200 BC: Hebrew

■ Atbash Cipher

- A simple monoalphabetic substitution cipher
- Originally developed for use with the Hebrew alphabet
- In the **Book of Yirmeyahu (Jeremiah)**, there are several words that have been enciphered through the use of the Atbash Cipher.
- Substitutes A with Z, B with Y, C with X, and so on.

Plaintext	A	B	C	D	E	F	G	H	I	J	K	L	M
Ciphertext	Z	Y	X	W	V	U	T	S	R	Q	P	O	N

Plaintext	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Ciphertext	M	L	K	J	I	H	G	F	E	D	C	B	A

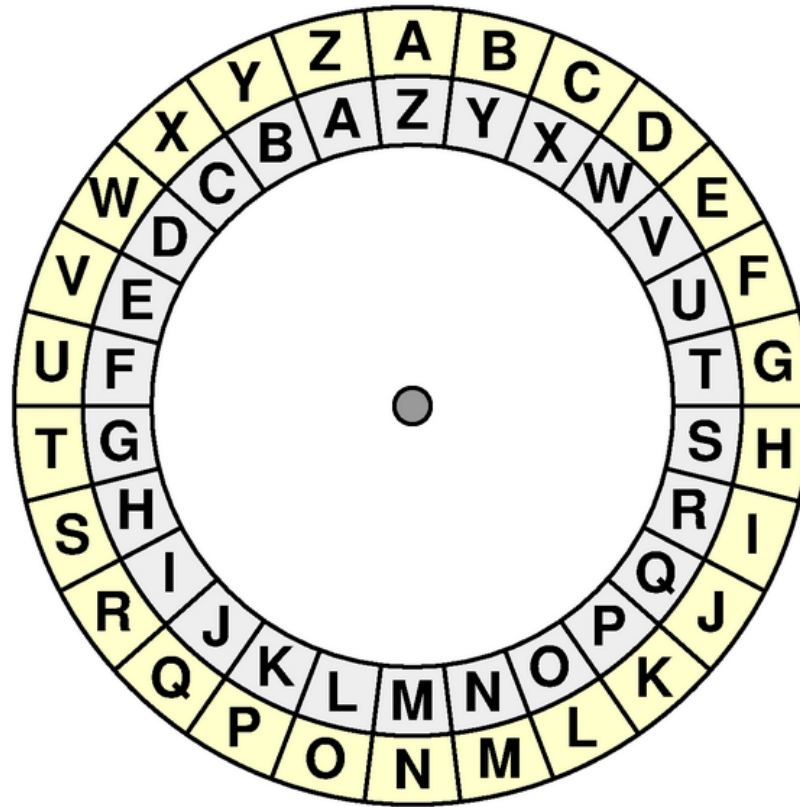


https://en.wikipedia.org/wiki/Book_of_Jeremiah

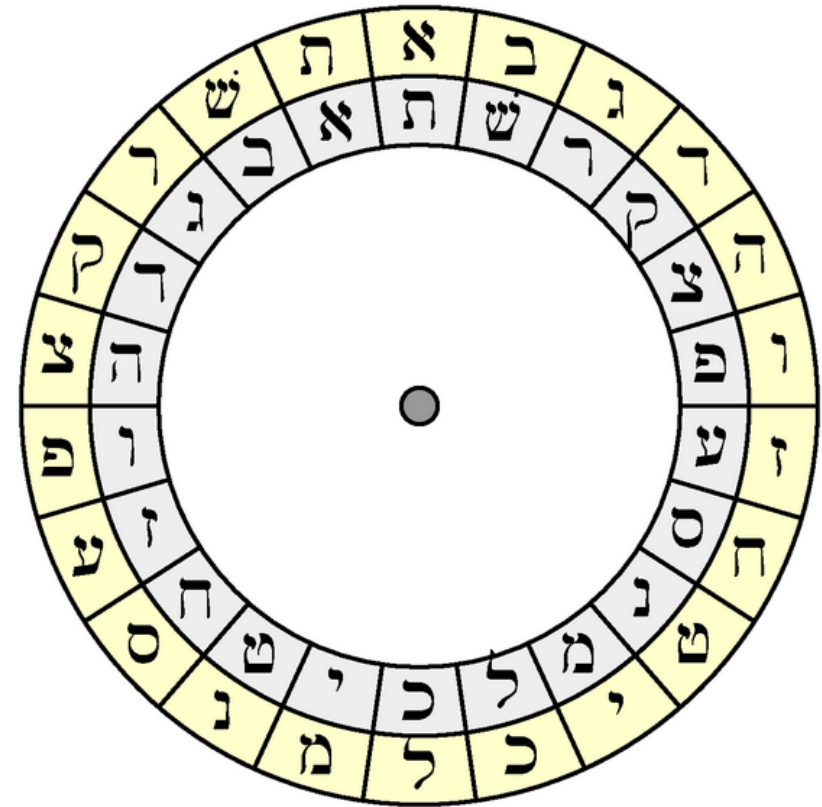
ANCIENT/HISTORICAL ENCRYPTION (CONT.)

500-200 BC: Hebrew

- **Atbash Cipher**



Atbash Wheel



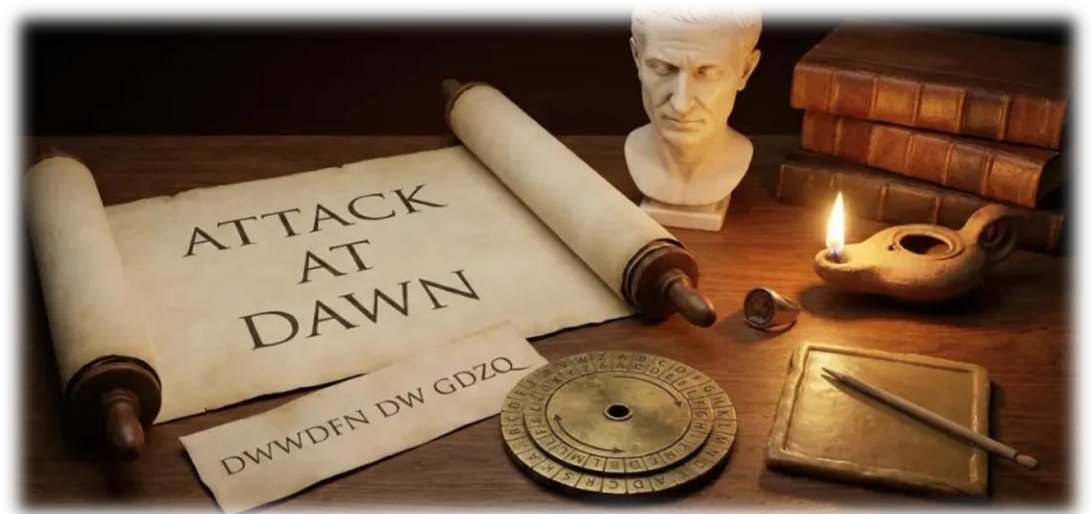
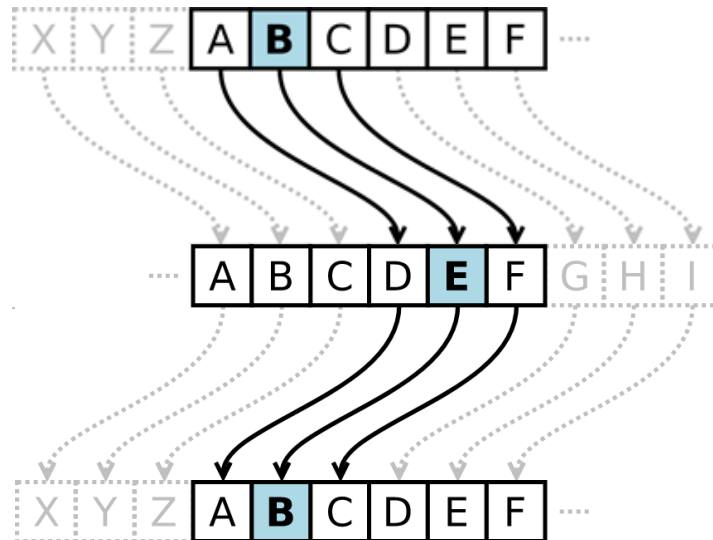
Original Atbash cipher(in Hebrew)

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

100 BC – 1st Century: Ancient Rome

■ Caesar Cipher

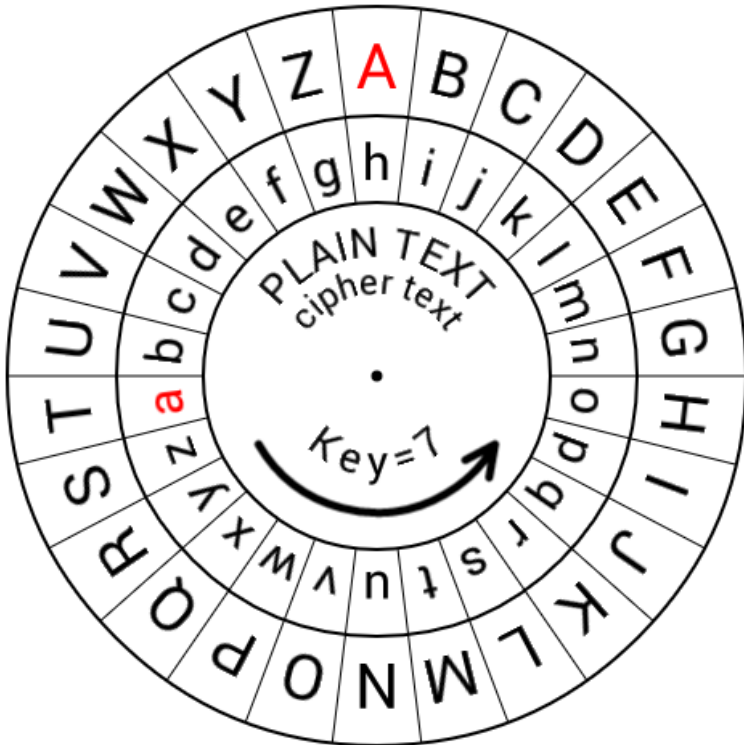
- Monoalphabetic Substitution
- A simple shift/rotate cipher used by Julius Caesar
- Replacing each letter with one a few positions (typically three) away in the alphabet



ANCIENT/HISTORICAL ENCRYPTION (CONT.)

100 BC – 1st Century: Ancient Rome

■ Caesar Cipher



Plaintext

Ciphertext

A	B	C	D	E	F	G	H	I	J	K	L	M
D	E	F	G	H	I	J	K	L	M	N	O	P

Plaintext

Ciphertext

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Q	R	S	T	U	V	W	X	Y	Z	A	B	C

HASHULHQFHLVWKHW
HDFKHURIDOOWKLQJV

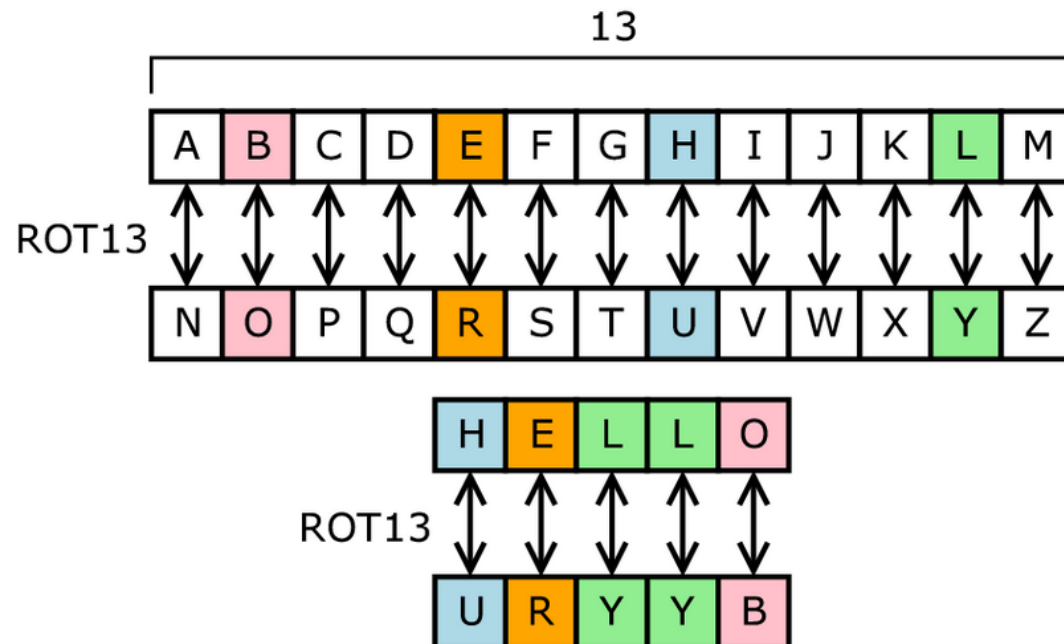
Can you decrypt it?

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

100 BC – 1st Century: Ancient Rome

■ ROT13 Cipher

- May be referred to **Rotate13**, **ROT-13** (*rotate by 13 places*)
- A special case of the Caesar cipher in which the shift is **always 13**.



RAPELCGNAQQRPELCG
What is the plaintext of this code?

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

100 BC – 1st Century: Ancient Rome

■ Caesar Cipher

- **Breaking a Caesar cipher** is straightforward if the encryption key is known, as one can simply reverse the shift applied to the plaintext.
- Furthermore, if an attacker knows the plaintext consists of English text, the cipher can be broken easily even without the key.
- By **testing various shift values**—ranging from $n=1$ and $n=2$ up to $n=25$ —and requiring a maximum of only 25 attempts (for the English language), the encrypted message can ultimately be deciphered.
- Try to decrypt the following two messages using the Caesar Cipher without knowing the key (each message uses a different key).

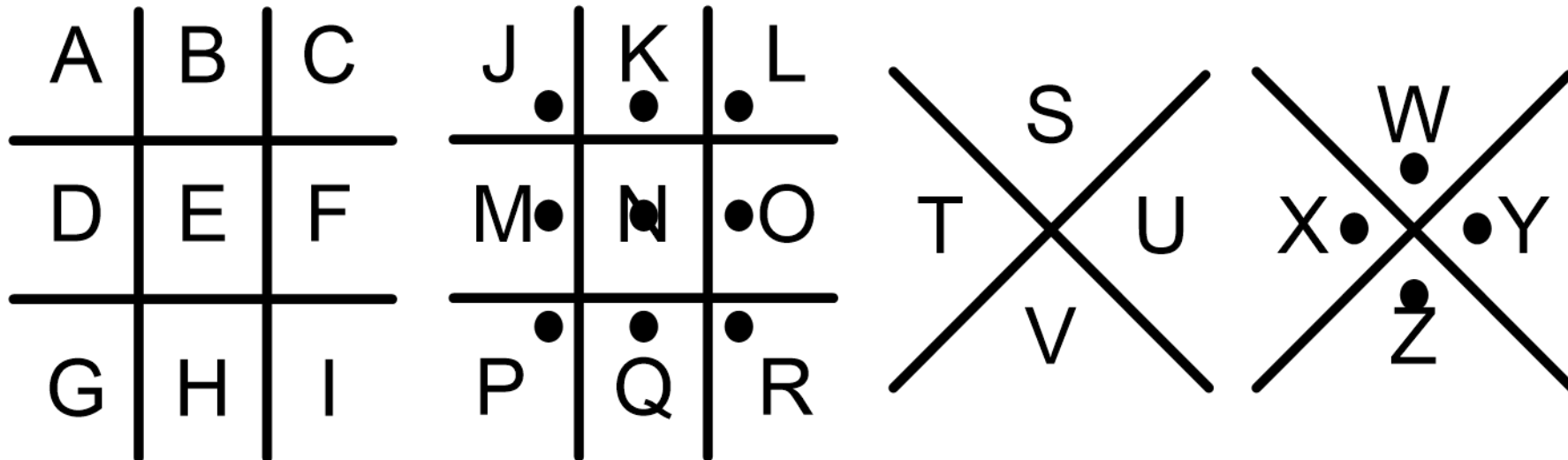
MYWOROBO OCZMZDNVOMVDOJMVHJIBPN

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

18th Century: The Freemasons

■ Pigpen Cipher

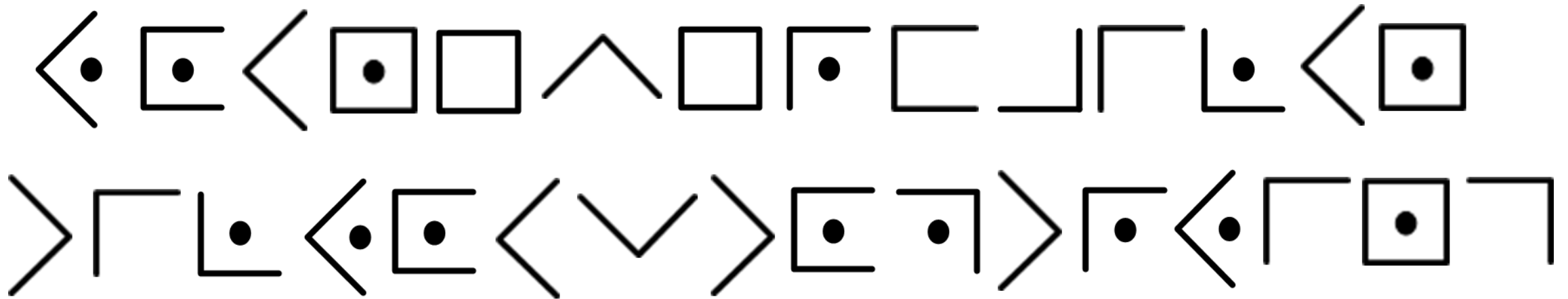
- Often called the **Masonic cipher**, **Freemason's Cipher**, **Tic-Tac-Toe Cipher**
- Widely use in their private records, documents, and on gravestones
- Use a simple geometric substitution code that replaces letters with fragments of a grid or X-shape



ANCIENT/HISTORICAL ENCRYPTION (CONT.)

18th Century: The Freemasons

■ Pigpen Cipher



ANCIENT/HISTORICAL ENCRYPTION (CONT.)

16th Century - Polyalphabetic Substitution

■ Vigenère Cipher

- Actually published by **Giovan Battista Bellaso**
- It was misattributed (wrongly give credit) to French cryptologist **Blaise de Vigenère**
- Use a table as a secret code; call this table **Vigenère Square** (Tabular recta)



Giovan Battista Bellaso



Blaise de Vigenère

		--PLAINTEXT--					
		A	B	C	D	...	Z
KEY	A	A	B	C	D	...	Z
	B	B	C	D	E	...	A
	C	C	D	E	F	...	B
	D	D	E	F	G	...	C
Y	:	:	:	:	:		:
	Z	Z	A	B	C	...	Y

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

16th Century - Polyalphabetic Substitution

■ Vigenère Cipher

- The Vigenère Square consists of the 26 letters of the English alphabet arranged sequentially both horizontally and vertically, forming a **26x26 grid**.
- It is a well-known form of encryption, valued for its simplicity and ease of implementation.
- In encryption, a **specific word** is designated to be paired with the available plaintext in order to generate ciphertext.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

16th Century - Polyalphabetic Substitution

■ Vigenère Cipher

- For example, let the plaintext be "**YOU WILL NEVER WALK ALONE**" and use the keyword "**BRITISH**", matching them character by character to align with the length of the plaintext. Then, use each pair to locate the corresponding column and row positions within the Vigenère table.

Plaintext	Y	O	U	W	I	L	L	N	E	V	E	R	W	A	L	K	A	L	O	N	E
Keyword	B	R	I	T	I	S	H	B	R	I	T	I	S	H	B	R	I	T	I	S	H
	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
Ciphertext	Z	F	C	P	Q	D	S	O	V	D	X	Z	O	H	M	B	I	E	W	F	L

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

16th Century - Polyalphabetic Substitution

■ Vigenère Cipher

- Or alternatively, each one shifted by one more letter with respect to the previous one. Hence, row number 1 represents a cipher alphabet with a Caesar shift of 1, row number 2 represents a cipher alphabet with a Caesar shift of 2, and so on.
- For example, if you want to encrypt the plaintext "SUBSTITUTION", the result will be

Plaintext	S	U	B	S	T	I	T	U	T	I	O	N
Ciphertext	T	W	E	W	Y	O	A	C	C	S	Z	Z

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
2	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
3	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
4	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
5	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
6	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
7	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
8	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
9	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
10	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
11	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
12	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
13	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
14	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
15	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
16	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
17	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
18	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
19	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
20	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
21	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
22	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
23	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
24	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
25	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
26	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

16th Century - Polyalphabetic Substitution

- **Vigenère Cipher**

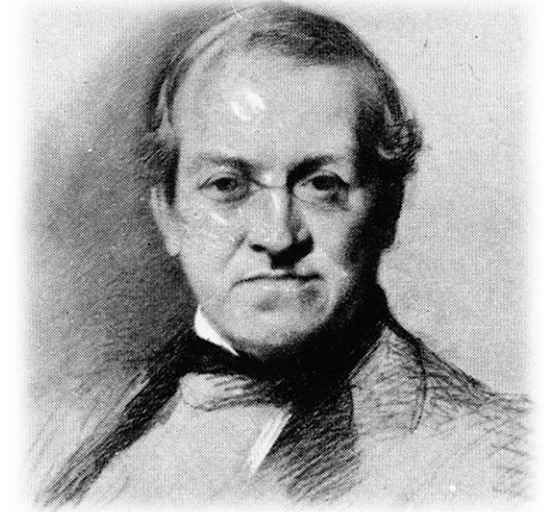
- Test decoding the message "DFQIHNOODRGVT" using the keyword "BOSTON".

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

18th - 19th Century - Polygraphic Substitution

■ Playfair Cipher

- Named after **Lord Playfair**, the cipher was actually invented by the English scientist and polymath **Sir Charles Wheatstone**
- Major milestone as the **first practical digraph substitution cipher** (encrypting pairs of letters instead of single letters)
- Construct a 5x5 key grid, populating it with characters—omitting specific letters like '**J**' or '**Q**' to ensure all other letters fit into the grid without duplication.
- For instance, using the key phrase "**PLAYFAIR EXAMPLE**".



P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

18th - 19th Century - Polygraphic Substitution

■ Playfair Cipher

- Take the original plaintext and divide the characters into pairs. If there are identical consecutive characters, insert an 'X' between them; if the final character does not form a complete pair, add a 'X'.
- For example, let the original message be:

HIDE THE GOLD IN THE TREE STUMP

- They can be divided into character pairs (digraphs) as follows:

HI DE TH EG OL DI NT HE TR EX ES TU MP

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

18th - 19th Century - Polygraphic Substitution

■ Playfair Cipher

- Compare the resulting pair of letters against the previously established key table based on the following conditions:
 - If the two letters are not in the same row or column, replace them by *swapping them diagonally*.
 - If the two letters are in the same column, replace each letter with the *one directly below it* (if a letter is at the bottom, replace it with the one at *the very top*).
 - If the two letters are in the same row, replace each letter with the *one directly to its right* (if a letter is at the far right, replace it with the one at *the far left*).

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

18th - 19th Century - Polygraphic Substitution

■ Playfair Cipher

HI DE TH EG OL DI NT HE TR EX ES TU MP

HI → **BM**

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

DE → **OD**

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

TH → **ZB**

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

18th - 19th Century - Polygraphic Substitution

■ Playfair Cipher

HI DE TH **EG** **OL** **DI** NT HE TR EX ES TU MP

EG → **XD**

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

OL → **NA**

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

DI → **BE**

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

18th - 19th Century - Polygraphic Substitution

■ Playfair Cipher

HI DE TH EG OL DI **NT HE TR** EX ES TU MP

NT → **KU**

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

HE → **DM**

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

TR → **UI**

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

18th - 19th Century - Polygraphic Substitution

■ Playfair Cipher

HI DE TH EG OL DI NT HE TR **EX ES TU** MP

EX → **XM**

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

ES → **MO**

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

TU → **UV**

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

18th - 19th Century - Polygraphic Substitution

- **Playfair Cipher**

HI DE TH EG OL DI NT HE TR EX ES TU MP

MP → **IF**

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

18th - 19th Century - Polygraphic Substitution

■ Playfair Cipher

- After the original text undergoes encoding based on character pairs, the resulting text is

H	I	D	E	T	H	E	G	O	L	D	I	N	T	H	E	T	R	E	X	E	S	T	U	M	P
B	M	O	D	Z	B	X	D	N	A	B	E	K	U	D	M	U	I	X	M	M	O	U	V	I	F

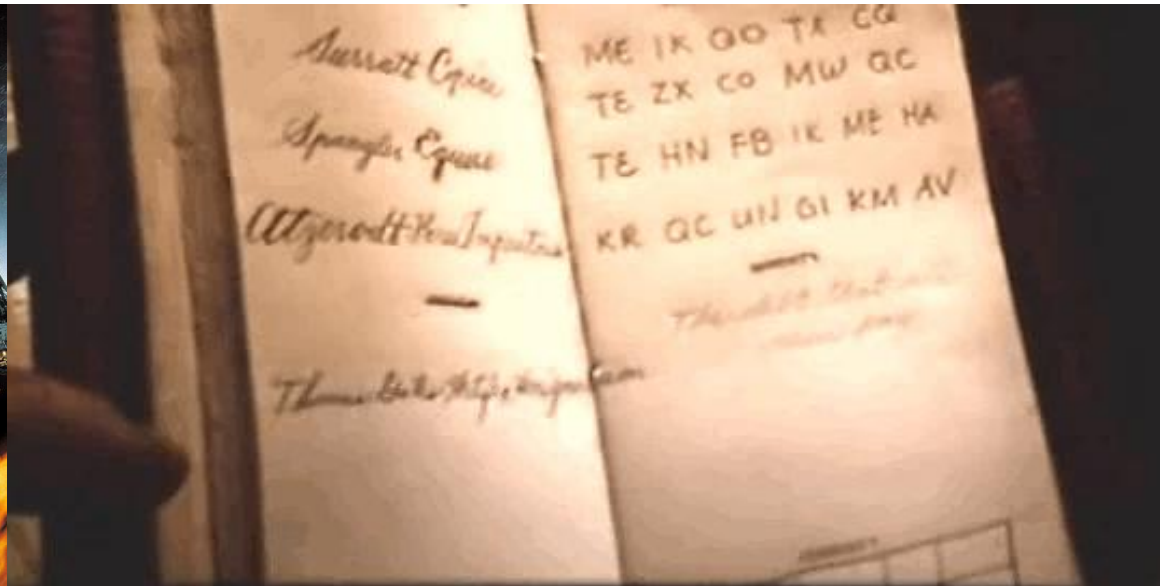
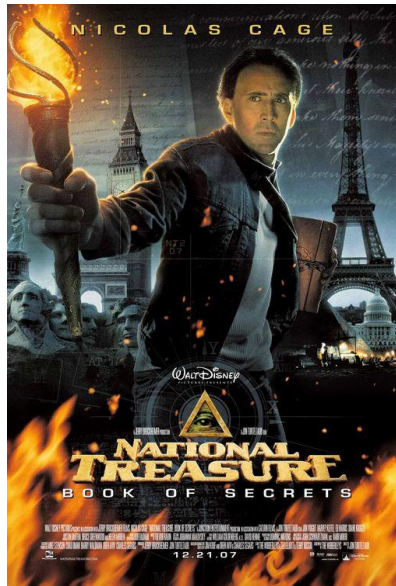
BMODZBXDNABEKUDMUIXMMOUIVIF

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

18th - 19th Century - Polygraphic Substitution

■ Playfair Cipher

- In the movie **National Treasure: Book of Secrets**, a message encrypted using the Playfair cipher is decoded. The ciphertext is: **ME IK QO TX CQ TE ZX CO MW QC TE HN FB IK ME HA KR QC UN GI KM AV**, and the keyword is **DEATH**. What is the plaintext message?



ANCIENT/HISTORICAL ENCRYPTION (CONT.)



Early 19th Century

■ Vernam Cipher

- It is a cryptographic system developed by **Gilbert Sandford Vernam** of AT&T Bell Labs during World War I.
- It employs a specific sequence of characters for encryption that is used **only once** and **never reused**; this sequence is known as "**one-Time Pad text**."
- Requirements for make one-time pad system **unbreakable**
 - The key must be at least as long as the plaintext
 - The key must be truly random
 - The key must never be reused in whole or in part
 - The key must be kept completely secret by the communicating parties

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

Early 19th Century

■ Vernam Cipher

The Vernam Cipher encryption process is as follows:

- Represent the **plaintext** as a **sequence of numbers** corresponding to *their positions in the English alphabet*.
- Generate a **One-Time Pad text**—a sequence of arbitrary characters (which need **not carry any meaning**) of the *same length as the plaintext*—and similarly convert these characters into numbers based on their alphabetical positions.
- **Add** the numerical values of the plaintext to the *numerical values* of the One-Time Pad text; if any resulting sum exceeds 26, subtract 26 from it.
- **Convert** the resulting numerical sequence back into characters to obtain the ciphertext.

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

Early 19th Century

■ Vernam Cipher

Plaintext	H	E	L	L	O
	7	4	11	11	14
One-Time Pad Text	X	M	C	K	L
	23	12	2	10	11
SUM	30	16	13	21	25
Mod 26	4	16	13	21	25
Ciphertext	E	Q	N	V	Z

Ciphertext	E	Q	N	V	Z
	4	16	13	21	25
One-Time Pad Text	X	M	C	K	L
	23	12	2	10	11
Ciphertext - Key	19	4	11	11	14
Mod 26	7	4	11	11	14
Plaintext	H	E	L	L	O

Once this encryption process is complete, this specific One-Time Pad text will not be reused; instead, the keyword will be *changed continuously and irregularly*, making it extremely difficult to guess the keyword.

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

Early 19th Century

■ Vernam Cipher

- If you decrypting the ciphertext "RRQNDLWDJRAL" using the key "PASYKXQMJCTN", what does the resulting plaintext read?

Ciphertext	R	R	Q	N	D	L	W	D	J	R	A	L
One-Time Pad Text	P	A	S	Y	K	X	Q	M	J	C	T	N
Ciphertext - Key												
Mod 26												
Plaintext												

ANCIENT/HISTORICAL ENCRYPTION (CONT.)

Early 19th Century

■ Vernam Cipher

Plaintext	Y	O	U	W	I	L	L	N	E	V	E	R	W	A	L	K	A	L	O	N	E
Keyword	B	R	I	T	I	S	H	B	R	I	T	I	S	H	B	R	I	T	I	S	H

Ciphertext